MEng Thesis Assessment Rubric

Design and Implementation [40%]

Level	System Modelling (15%)	Level	System Development (15%)	Level	Use of literature to support development (10%)
Fail	Section missing, or unable to meet the next rubric level.	Fail	Section missing, or unable to meet the next rubric level.	Fail	No use of literature, or unable to meet the next rubric level.
Unsatisfactory	Methods used maybe overly simplistic and/or not suitable for the intended application. Justification is poor or missing.	Unsatisfactory	Development is significantly lacking, and it is clear that little actual effort has been expended on the project. There is no bespoke work, instead the project is comprised of off-the-shelf solutions.	Unsatisfactory	There is some attempt to reference where supporting material is used but it is unclear what aspects of the project are original work and which are from external material. No considerations of the limitations of referenced material.
50% (Pass)	SStudent demonstrates the use of simulation and/or analytical methods to model/design their project and makes an attempt at discussing the limitations. The methods / tools / techniques that are used are justified with respect to the project specification. Analysis is clearly linked to the project specification and beyond direct step-by-step following of reference materials (e.g. software tutorials). LO8. Select and apply appropriate engineering tools to model and solve complex electrical and electronics engineering problems. *M3. Select and apply appropriate computational and analytical techniques to model complex problems, recognising the limitations of the techniques employed.*	50% (Pass)	Student demonstrates ability to take designed and modelled system into a 'real world' environment (note this may be through an actual build or a further refined simulation). Expected challenges from this process are discussed and compensated for. The development makes use of bespoke methods created by the student to meet the project specifications. LO9. Apply comprehensive set of engineering principles to solve complex problems, where much of the knowledge is at either at the forefront of the topic or emerging. *M1. Apply a comprehensive knowledge of mathematics, statistics, natural science, and engineering principles to the solution of complex problems. Much of the knowledge will be at the forefront of the particular subject of study and informed by a critical awareness of new developments and the wider context of engineering.*	50% (Pass)	Student makes clear referenced use of sources to support their project development. May be from poor sources or focussing on a single type (e.g. website). IEEE/IET style is followed by-in-large correctly. Limitations of the referenced sources are considered with respect to the project needs and specification requirements. LO1. Identify and critically evaluate the literature to solve a complex electrical and electronic engineering problem. *M4. Select and critically evaluate technical literature and other sources of information to solve complex problems.*
+++	Clear discussion of the limitations and/or trade-offs of the techniques with reference to the project specification, application area, wider context, and project originality.	+++	Student has demonstrated competency with a range of multidisciplinary tools throughout the development of their project.	++++	Student clearly identifies how they have adopted and/or adapted the referenced material to add value to their project and the originality they have contributed.
++	The methods / tools / techniques that are used are justified with respect to the project specification as well as application area, resources, and wider context.	++	The development shows a clear progression of their methodology adhering to engineering principles. All changes from the original methodology are both explained and justified well.	++	All materials are from high quality sources (e.g. manufacturer, peer reviewed, textbook) and a range of types (datasheets, app notes, tutorials, code samples, etc).
+++++	Demonstrate application of the methods / tools / techniques to successfully address the project specifications.	+++	Clear description of sub-system testing that suitably validates individual components before integration as a complete system.	++++	Referencing is used consistently throughout the implementation section to support both the design, simulation, and (where applicable) the practical implementation.
		++	The originality of the project is indicated and demonstrated through the development		

Final System Testing and Validation [30%]

Level	Specification Validation (20%)	Level	Presentation of testing setup and results (10%)
Fail	Section missing, or unable to meet the next rubric level. Results are provided with no context.	Fail	No results have been presented.
Unsatisfactory	Results are not used to validate specification or compared with expectations.	Unsatisfactory	Some results but the chosen format or order is unintelligible / confusing.
50% (Pass)	Student captures results from their project and uses these to validate against the specification list. Analysis of results may be rudimentary (e.g. only stating visible features/trends). Results are compared with expectations which may be from modelling, literature, or component/method performance differences. LO11. Interpret and critically evaluate available data to substantiate a conclusion based on core principles of mathematics and engineering. *M2. Formulate and analyse complex problems to reach substantiated conclusions. This will involve evaluating available data using first principles of mathematics, statistics, natural science and engineering principles, and using engineering judgment to work with information that may be uncertain or incomplete, discussing the limitations of the techniques employed..*	50% (Pass)	Testing setup and results are presented using suitable figure types (image, bar, scatter, pie, etc). These are clear with suitable labels, readable axes and correct units etc. Logical ordering of results structured intuitively with the discussion sections. LO6. Communicate the solution to a complex engineering problem to a technical audience. *M17. Communicate effectively on complex engineering matters with technical and non-technical audiences, evaluating the effectiveness of the methods used.*
++	Results are compared with expectations and a realistic pass/fail criterion is given. Unexpected results are explained well with suitable attempt to validate the explanation.	+++	Use of a range of figure/table types as appropriate for the data being displayed, with suitable axis scales, units, and labels.
+++	Detailed statistical/quantitative analysis of results are carried out to ensure confidence in accuracy and repeatability.	+++	Clarity of comparison, highlighting/indication of key aspects and features for further discussion (e.g. subfigures, arrows, circles, etc).
+++	Every point in the specification list is validated successfully. Where specification points are not met there is clear explanation why with analysis of the impact on the overall functionality of the final system.	++++	Testing setup details the test conditions and procedures, with justification.
++	Results are discussed with respect to other work in the literature and used to support the originality of the project.		

Key

Achievement criteria
Module learning outcome
IET AHEP4 competency

None	Some	Most	Fully met	Exceeded

Conclusion [25%]

Level	Consideration of system within the wider context (10%)	Level	Reflection on Management (15%)
Fail	Section missing, or unable to meet the next rubric level.	Fail	Section missing, or unable to meet the next rubric level.
Unsatisfactory	Student summarises their work but has limited or no discussion of the wider background. There is no consideration of how the project could be further developed.	Unsatisfactory	Very limited reflection with no lessons learnt.
50% (Pass)	Student reviews their final system with respect to the wider background. This can include consideration of health & safety, diversity, inclusion, cultural, societal, environmental, and commercial matters, codes of practice, and industry standards. Detailed considerations are made to how the project would next be developed. LO10. Design solutions to complex a problem that evidences some originality that meets a specified requirement. *M5. Design solutions for complex problems that evidence some originality and meet a combination of societal, user, business and customer needs as appropriate. This will involve consideration of applicable health & safety, diversity, inclusion, cultural, societal, environmental and commercial matters, codes of practice and industry standards.*	50% (Pass)	Student will reflect upon the progress of their project in terms of the project management. They must have an included an appendix containing copies of monthly progress review pro-forma's, which they refer to. The success of the risk mitigation strategies are evaluated. Lessons learnt will be discussed but may be unclear how these would relate to future projects. LO7. Critically reflect on the process of solving a complex engineering problem and outcomes achieved. *M18. Plan and record self-learning and development as the foundation for lifelong learning/CPD.*
+++	Detailed considerations of how the project would next be developed includes details of resource (cost, time, etc) requirements, and covers both immediate and longer-term goals. These are assessed with respect to other work in the field.	+++	Project management skills that have been developed are clearly described in a way that is suitable for future engineering projects.
++++	Assessment of the final project demonstrates a strong understanding of how it fits within a wide societal context. Including consideration of health & safety, diversity, inclusion, cultural, societal, environmental, and commercial matters, codes of practice, and industry standards; all as applicable to the project and its application area(s). The potential impact of the project is extrapolated and quantified with respect to these contexts.	+++	Reflections on progress and any actions undertaken to keep the project on track supported by reference to the review documents.
+++	Summary of the current state of the project including full listing of the status of every specification point and the overall project aim. Where points are not met these are accompanied by a suitable explanation and justification, where everything is met the challenge posed by the project is clearly explained. Student describes how the final project meets the originality goals outlined in their planning stage.	++++	Suitable learnings that can be applied to planning future projects are detailed with acknowledgement of why this was not done in the original plan.

Communication Quality and Overall Completeness of Document (5%)	
Level	Communication Quality and Overall Completeness of Document (5%)
Fail	Section missing, or unable to meet the next rubric level.
Unsatisfactory	The thesis is incomplete and/or poorly presented or written.
50% (Pass)	The thesis is given as a complete document that would allow another person, with a technical background but unfamiliar with the project, to continue or repeat the work. The overall document follows a suitable layout and has an acceptable presentation with expected document features (section headings, page numbers, contents page). Inclusion of tables of figures, tables, abbreviations, and symbols where applicable. Correct use of cross-referencing throughout the document. LO6. Communicate the solution to a complex engineering problem to a technical audience. *M17. Communicate effectively on complex engineering matters with technical and non-technical audiences, evaluating the effectiveness of the methods used.*
+++	Consistent use of styles throughout the thesis including headings, page layout, captioning, etc.
+++	Overall quality of written communication is excellent and concise; with correct grammar, spelling, etc.
++++	Figures are used to present the work in the most intuitive and engaging way to the reader, avoiding large blocks of text.

Moderator/Supervisor assessments - separated

Level	Dissertation Defence (10% - Moderator)	Level	Professionalism and Self Management (10% - Supervisor)
Fail	Student did not attend, or unable to meet the next rubric level..	Fail	Little to no attendance at meetings, no evidence of progress between meetings.
Unsatisfactory	Student is unable to answer simple questions about the project..	Unsatisfactory	Student demonstrated limited aspects of self-management during the project and required micromanagement to make progress. Meetings often missed with no notice or explanation.
50% (Pass)	The work of the project (including aim) is well explained with suitable demonstration (e.g. live, video, results discussion) of the final system. Student clearly states which specification points have been met and discusses any deficiencies. Basic questions about the students experience of the project are handled well. Delivery is clearly practiced and performed with genuine confidence, enthusiasm, and professionalism. LO6. Communicate the solution to a complex engineering problem to a technical audience.. *M7. Communicate effectively on complex engineering matters with technical and non-technical audiences..*	50% (Pass)	Student was generally self-managed during the project and did not require much, if any, micromanagement. Meetings were attended in line with the expectations of the supervisor (including provision of agendas where expected). LO6. Communicate the solution to a complex engineering problem to a technical audience.. *M17. Communicate effectively on complex engineering matters with technical and non-technical audiences..*
++++	Questions about the project specifics (including methods applied, reasoning, validity of results, originality of work) are expertly handled.	+++++	Purely self-motivated with a consistent high level of effort applied throughout the project.
++	Questions about their personal development and experience working on the project are expertly handled.	+++	Student has a professional approach to supervisor contact and arrives to meetings prepared to maximise value of time spent.
++	Questions reflecting on their planning process and how the project went are expertly handled.	++	Student self-directed the originality and creativity within the project?
++	Reflection on presentation method – prompting questions. "Why did you choose the method you did to present your project, what factors did you consider?"		